

```
In [1]: # data manipulation
import pandas as pd
import numpy as np

#import attricion data
df=pd.read_csv('C:\\Users\\Public\\Desktop\\DAT-430\\HR Attrition Data.csv')
```

```
In [2]: #inspect the data
df.head()
```

Out[2]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education
0	38	Yes	Travel_Rarely	1180	Research & Development	29	1
1	49	No	Travel_Frequently	279	Research & Development	8	1
2	28	Yes	Non-Travel	1366	Research & Development	24	2
3	33	No	Travel_Frequently	1392	Research & Development	3	4
4	27	No	Travel_Rarely	591	Research & Development	2	1

5 rows × 35 columns



```
In [3]: df.isnull().sum()
```

```
Out[3]: Age                                0
Attrition                                0
BusinessTravel                           0
DailyRate                                0
Department                               0
DistanceFromHome                         0
Education                                0
EducationField                            0
EmployeeCount                             0
EmployeeNumber                           0
EnvironmentSatisfaction                   0
Gender                                    0
HourlyRate                               0
JobInvolvement                           0
JobLevel                                 0
JobRole                                  0
JobSatisfaction                           0
MaritalStatus                            0
MonthlyIncome                             0
MonthlyRate                              0
NumCompaniesWorked                       0
Over18                                    0
OverTime                                  0
PercentSalaryHike                         0
PerformanceRating                        0
RelationshipSatisfaction                   0
StandardHours                             0
StockOptionLevel                         0
TotalWorkingYears                        0
TrainingTimesLastYear                     0
WorkLifeBalance                           0
YearsAtCompany                            0
YearsInCurrentRole                        0
YearsSinceLastPromotion                   0
YearsWithCurrManager                      0
dtype: int64
```

```
In [4]: df.info()
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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   Age                                    1470 non-null   int64
 1   Attrition                             1470 non-null   object
 2   BusinessTravel                         1470 non-null   object
 3   DailyRate                             1470 non-null   int64
 4   Department                             1470 non-null   object
 5   DistanceFromHome                      1470 non-null   int64
 6   Education                             1470 non-null   int64
 7   EducationField                         1470 non-null   object
 8   EmployeeCount                         1470 non-null   int64
 9   EmployeeNumber                        1470 non-null   int64
10   EnvironmentSatisfaction               1470 non-null   int64
11   Gender                                1470 non-null   object
12   HourlyRate                           1470 non-null   int64
13   JobInvolvement                       1470 non-null   int64
14   JobLevel                             1470 non-null   int64
15   JobRole                              1470 non-null   object
16   JobSatisfaction                      1470 non-null   int64
17   MaritalStatus                       1470 non-null   object
18   MonthlyIncome                       1470 non-null   int64
19   MonthlyRate                         1470 non-null   int64
20   NumCompaniesWorked                  1470 non-null   int64
21   Over18                              1470 non-null   object
22   OverTime                             1470 non-null   object
23   PercentSalaryHike                   1470 non-null   int64
24   PerformanceRating                   1470 non-null   int64
25   RelationshipSatisfaction             1470 non-null   int64
26   StandardHours                       1470 non-null   int64
27   StockOptionLevel                    1470 non-null   int64
28   TotalWorkingYears                   1470 non-null   int64
29   TrainingTimesLastYear               1470 non-null   int64
30   WorkLifeBalance                     1470 non-null   int64
31   YearsAtCompany                      1470 non-null   int64
32   YearsInCurrentRole                  1470 non-null   int64
33   YearsSinceLastPromotion             1470 non-null   int64
34   YearsWithCurrManager                1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

```
In [5]: #remove columns
df.drop(['DailyRate',
        'EmployeeCount',
        'EmployeeNumber',
        'Over18'], axis=1, inplace=True)
```

```
In [6]: df.describe()
```

```
Out[6]:
```

	Age	DistanceFromHome	Education	EnvironmentSatisfaction	HourlyRate
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	9.192517	2.912925	2.721769	65.891156
std	9.135373	8.106864	1.024165	1.093082	20.329428
min	18.000000	1.000000	1.000000	1.000000	30.000000
25%	30.000000	2.000000	2.000000	2.000000	48.000000
50%	36.000000	7.000000	3.000000	3.000000	66.000000
75%	43.000000	14.000000	4.000000	4.000000	83.750000
max	60.000000	29.000000	5.000000	4.000000	100.000000

8 rows × 23 columns

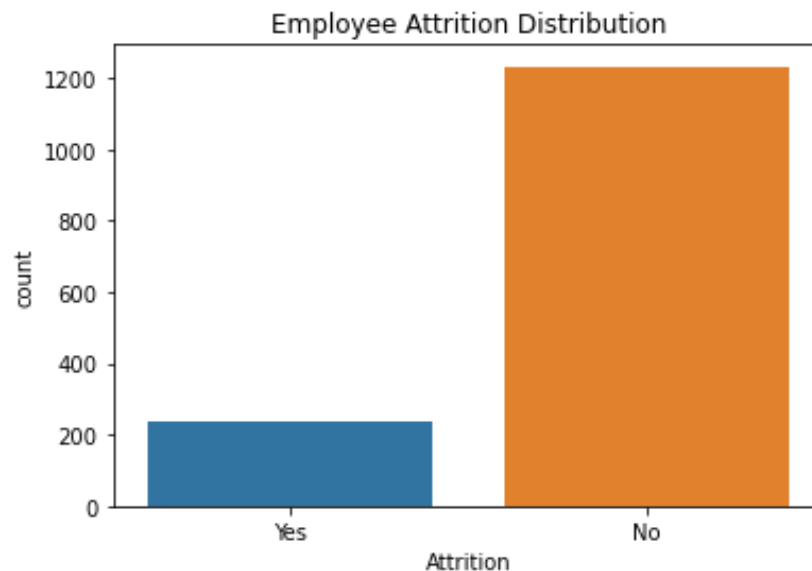


```
In [7]: #understanding attrition distribution
df["Attrition"].value_counts(normalize=True)
```

```
Out[7]: No      0.838776
Yes       0.161224
Name: Attrition, dtype: float64
```

```
In [8]: import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(data=df, x='Attrition')
plt.title('Employee Attrition Distribution')
plt.show()
```

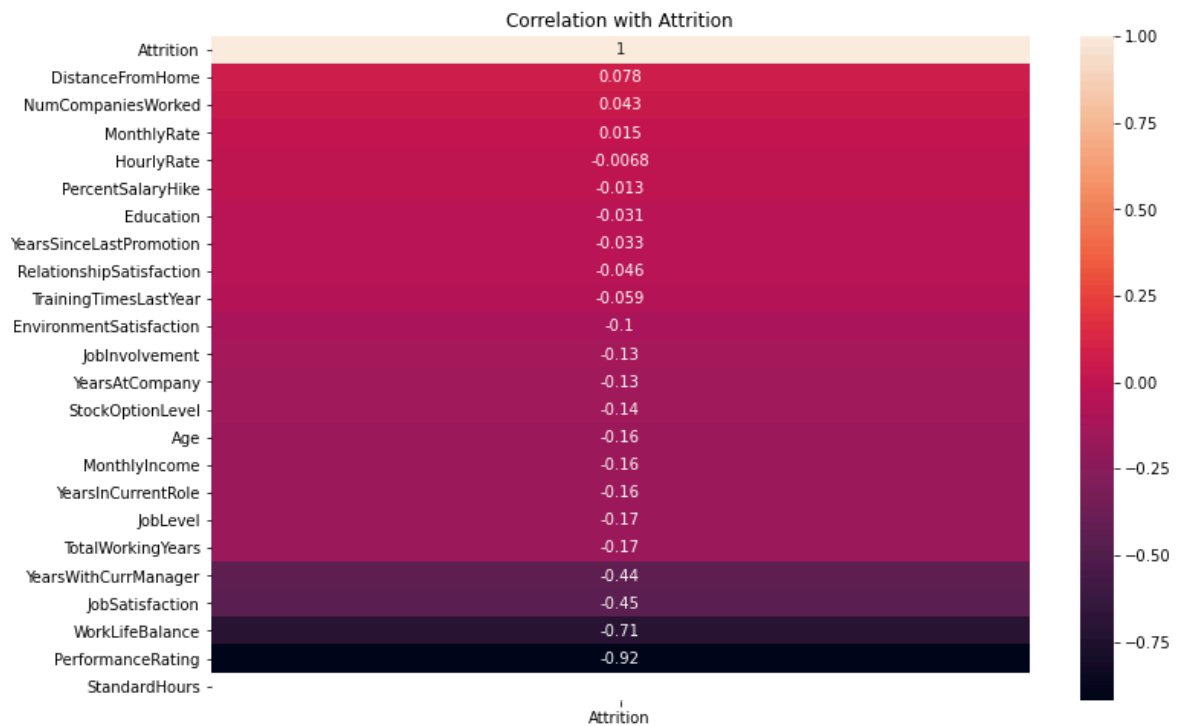


```
In [9]: #changing Attrition from object to numbers
df['Attrition']=df['Attrition'].map({"No":0,"Yes":1})
```

```
In [10]: import matplotlib.pyplot as plt

corr=df.corr()
plt.figure(figsize=(12,8))
sns.heatmap(corr[['Attrition']].sort_values(by='Attrition', ascending=False), annot=True)
plt.title('Correlation with Attrition')
plt.show
```

```
Out[10]: <function matplotlib.pyplot.show(*args, **kw)>
```



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In [11]: #checking unique value for later binarization
print("BusinessTravel:", sorted(df['BusinessTravel'].unique()))
print("Department:", sorted(df['Department'].unique()))
print("EducationField:", sorted(df['EducationField'].unique()))
print("Gender:", sorted(df['Gender'].unique()))
print("JobRole:", sorted(df['JobRole'].unique()))
print("MaritalStatus:", sorted(df['MaritalStatus'].unique()))
print("OverTime:", sorted(df['OverTime'].unique()))
```

```
BusinessTravel: ['Non-Travel', 'Travel_Frequently', 'Travel_Rarely']
Department: ['Human Resources', 'Research & Development', 'Sales']
EducationField: ['Human Resources', 'Life Sciences', 'Marketing', 'Medical', 'Other', 'Technical Degree']
Gender: ['Female', 'Male']
JobRole: ['Healthcare Representative', 'Human Resources', 'Laboratory Technician', 'Manager', 'Manufacturing Director', 'Research Director', 'Research Scientist', 'Sales Executive', 'Sales Representative']
MaritalStatus: ['Divorced', 'Married', 'Single']
OverTime: ['No', 'Yes']
```

```
In [12]: #binarization
df['OverTime'] = df['OverTime'].map({'No': 0, 'Yes': 1})
df['BusinessTravel'] = df['BusinessTravel'].map({"Non-Travel":0, "Travel_Rarely":1, "Travel_Frequently":2})
df['Department']=df['Department'].map({'Human Resources':0, 'Research & Development':1, 'Sales':2})
df['JobRole']=df['JobRole'].map({'Healthcare Representative':0, 'Human Resources':1, 'Laboratory Technician':2, 'Manager':3, 'Manufacturing Director':4, 'Research Director':5, 'Research Scientist':6, 'Sales Executive':7, 'Sales Representative':8})
df['EducationField']=df['EducationField'].map({'Human Resources':0, 'Life Sciences':1, 'Marketing':2, 'Medical':3, 'Other':4, 'Technical Degree':5})
df['Gender']=df['Gender'].map({'Female':0, 'Male':1})
df['MaritalStatus']=df['MaritalStatus'].map({'Divorced':2, 'Married':1, 'Single':0})
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